

Literature Survey and Reuse Decisions

SVG Patch Lab - Research Working Package

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Updated 13 September 2026. This supplements the earlier graphics survey with the user’s clarified focus: engineering problems solved using physics and FEM.

1 Conclusion

The broad idea already exists, including FEM agents, solver verification, and fine-tuning on verified FEM programs. A paper cannot claim those combinations as new. The candidate extension is maintaining numerical and semantic correctness in the final editable engineering drawing, including after physical and presentation edits. This remains a hypothesis to test against existing simulation and graphics systems; the search does not establish unique novelty.

2 Closest primary sources

Work	What overlaps	What to reuse or compare
MechAgents , Ni & Buehler, Extreme Mechanics Letters, 2024	LLM agents formulate, execute and correct FEM elasticity programs.	A historical solver-agent baseline; published Colab notebooks require older FEniCS/AutoGen dependencies. Do not call a modern reimplementaion an exact reproduction. Paper , author code .
MCP-SIM , npj Artificial Intelligence, 2026	Language-driven simulation through tools and solver workflows.	Closest earlier simulation orchestration comparison from the first survey; runnable-release limitations are recorded there. Paper , code .
FEABench , Mudur et al., arXiv 2025; paper notes NeurIPS 2024 workshop acceptance	End-to-end natural-language engineering problems solved through COMSOL APIs, with iterative tool feedback.	Reuse the distinction between an executable solver call and a quantitatively correct solution. COMSOL availability is a dependency for faithful reproduction. Paper and linked code .
FEM-Bench , Mohammadzadeh et al., December 2025, revised May 2026	FEM function generation and test generation against reference and deliberately broken implementations.	Selected for immediate code reuse. MIT-licensed tasks provide auditable numerical anchors. Its 33-task results are not directly comparable to our SVG editing scores. Paper , repository .
ALL-FEM , Deotale et al., CNAME 457:118985, 2026	Fine-tuned models generate FEniCS code in an agentic, feedback-driven workflow. The authors report over 1,000 verified scripts, models from 3B to 120B, and 39 engineering benchmarks.	The most direct precedent for the proposed training component. Training a small solver agent alone is not a new contribution. Its reported code-level success is not a final-drawing correctness metric. Paper , author project and journal citation .
PDEAgent-Bench , May 2026	645 PDE instances across 11 families, with DOLFINx, Firedrake and deal.II tracks; numerical accuracy and efficiency beyond mere execution.	Use its calibrated-reference philosophy and PDE families when expanding beyond elementary anchors. Keep benchmark test cases out of training. Paper , code .

Work	What overlaps	What to reuse or compare
OASiS / Open FEM Agent , 2026 software	MCP access to multiple FEM solvers, numerical verification, convergence studies and visualization.	A current open-source orchestration baseline; its scikit-fem backend matches our local solver ecosystem. This is software prior art, not an assumed peer-reviewed performance result. We have inspected its documentation, not reproduced its full agent evaluation. Repository and citation, archived earlier release.
VFEAgent , May 2026, revised August 2026, preprint	Images and text are converted into structured FEA specifications, then simulation code with verification and debugging.	Directly relevant to CVPR positioning: multimodal engineering-drawing-to-FEA workflows already exist. Compare perception, physical specification and final artifact separately. Paper.
ModSolAgent , IEEE Transactions on Industrial Informatics, June 2026	Abaqus script generation with retrieval and iterative verification, plus distilled fine-tuning data.	Another direct precedent for training engineering code models. Abaqus licensing and released assets must be checked before reproduction. Publisher.

The prior [graphics survey](#) remains relevant for Penrose, scientific SVG generation and diagram editing. The engineering papers establish the solver side; graphics papers establish the editable-artifact side. Their combination still needs a precisely defined, experimentally justified contribution.

3 What has actually been built on top of prior work

The vendored source snapshot **vendor/FEM-bench** is pinned to commit d370aef5dbe023b0b86eb3548d24d11fd69328ee; its MIT license is retained. The new scripts/fem_bench_svg_extension.py directly imports the author’s unchanged FEM_1D_linear_elastic_CC0_H0_T0 reference function.

The bridge passes both upstream reference tests, solves 12 generated axial-bar cases, and performs 12 additional FEM solves after a 50% load increase. It writes displacements, reactions, provenance IDs and editable SVG plots for both states. All 24 solves agree with the analytic bar solution to below 2×10^{-18} m in this run; maximum relative force imbalance is below 4.2×10^{-14} . These are elementary numerical checks, not evidence of general engineering capability. A stale old tip displacement would be 33.3% low relative to the edited-load solution.

This is a working integration anchor. It does not reproduce the full FEM-Bench leaderboard, train a model on FEM-Bench, or demonstrate a new visual reasoning result. No upstream agent framework has been silently relabeled as ours. Results are in **runs/fem-bench-svg-anchor/**.

Separately, scripts/robin_fem_reference.py uses scikit-fem to implement the weak form for the exact historical convective-plate problem. It checks an analytic Robin anchor, discrete heat balance and five mesh resolutions. This is an independent FEM implementation, not a reproduction of MechAgents or ALL-FEM. See the [Astra recheck](#).

4 Next contribution to test

Use a structured physical specification and solver-produced numerical record as the source for an editable drawing. A physical change must cause an actual new solve; a presentation change must preserve the numerical claim and visible meaning. Measure errors after the final SVG is rendered and edited, not only in a solver array or API response.

The main comparison should include direct Astra SVG generation, a published solver-agent workflow, deterministic solver plotting, unrestricted solver-informed SVG editing, and protected numerical/semantic editing. Equalize available physical information and count every attempted case. Test real geometry and material changes, not only load scaling. Include human-written requests, multi-step edits, quantity/unit/legend errors, and visual layout quality. Training is justified only after a stronger task shows a gap that simple prompting or a deterministic parser does not already close.

Prioritize well-posed Robin heat transfer, finite-domain elasticity with explicit material/support/load assumptions, and electrostatics with a stated dimensional and far-boundary model. A sharp-corner stress singularity has no finite mesh-independent peak, so it is a disclosure/identifiability test unless a fillet and load/support model are specified. Reference uncertainty and singular regions must be declared before model scoring.

For CVPR, a language-only code router is insufficient evidence. The study needs an actual visual or graphics contribution and comparisons to multimodal FEA and scientific diagram-editing work. The completed A100 pilot is useful infrastructure; it is not the main FEM experiment.

5 Graphics and adjacent prior art

Work	Verified overlap	How this project could extend it
MechAgents , Ni and Buehler, preprint 2023; Extreme Mechanics Letters 2024	Agents write, execute, and correct FEM code for elasticity. Its examples explicitly include plotting the wrong stress quantity and subsequently correcting it.	Reuse its elasticity tasks and solver-agent baseline; automatically check that the displayed quantity and contours agree with the solution, including after user edits. Paper , author code and Colab notebooks .
MCP-SIM , Park, Moon and Ryu, npj Artificial Intelligence, January 2026	Natural-language specifications become simulations and explanations through clarification, execution, and correction. Its reported twelve-task success is specific to that evaluation.	Use its workflow as a named simulation baseline. Add explicit solver-to-SVG provenance and post-edit verification. Paper , author repository .
SSVG-Bench / LOOP , <i>Towards the Generation of Structured Scientific Vector Graphics with Large Language Models</i> , inspected ICLR 2026 submission	Scripted structural evaluation for plane geometry and molecular diagrams, spanning TikZ, SVG and EPS; reasoning-oriented prompting. The accessed manuscript is labeled under review; acceptance and released code were not verified.	Extend structural checks to numerical PDE fields and edit sequences; do not claim to invent non-visual correctness metrics for scientific diagrams. Manuscript .
Penrose , SIGGRAPH 2020	Separates mathematical meaning and visual styling, uses constraints for layout, and exports SVG.	Borrow the separation of semantics from layout. Physical contours must be computed and bound to the solution; layout constraints alone do not establish PDE accuracy. Author-hosted paper , code .
AutoFigure-Edit , ACL system demonstrations 2026	Generates editable scientific illustrations with reference-guided styling and an SVG editing interface.	Strong available graphics baseline: test whether its edits preserve solver-derived geometry, field labels and units. Published paper , author code .
SciDiagramEdit , July 2026 preprint	Edits scientific vector sources using instructions derived from paper revisions; improves an editor through skill evolution over execution traces.	Closest editing-learning comparison. Add numerical verification to the editing objective. Its skill evolution is not the same as weight fine-tuning. No official code release was verified in this audit. Paper .
SVGEditBench V2 , February 2025 preprint	Instruction-based SVG editing on emoji-derived examples, evaluated with visual, semantic and geometric metrics.	Retain it as a general editing control, and add physical-claim checks on engineering cases. It is not a physics benchmark. Paper .
FMforME , <i>A Three-Layer Runtime Constraint Verification Framework...</i> , Applied Sciences 2026	Validates structured CAD specifications and feeds detected defects back to the LLM. Its stated future work includes extending through FEA solving.	Compare specification validity with actual solved-field fidelity. Do not copy its domain-specific engineering rules as universal physical laws. Article .

Work	Verified overlap	How this project could extend it
SciForma , July 2026 preprint	Scientific methodology-diagram generation with component, arrow and text evaluation and multidimensional preference training.	Borrow the idea that all required correctness dimensions must pass. Its released renderer is a diffusion image model, so it is an adjacent visual baseline rather than a native SVG/PDE implementation. Paper , author repository .
Text2PDE , October 2024 preprint	Text-conditioned latent-diffusion physics simulation.	Relevant if extending into learned simulation, but much broader than the current need for trustworthy SVG export and editing. Paper .

6 Public-code feasibility

The inspected MCP-SIM main tree has six agent modules, a README and an MIT license. Several modules import `utils.setup_logger`, but `utils.py` is absent from that tree. A complete orchestration entry point and dependency manifest were also absent. Its executor runs generated Python and scans output for error patterns; that module does not itself establish numerical correctness. The release needs integration work before it can be treated as a reproducible baseline. This observation concerns the public code, not a claim that the paper’s private experiments were invalid. [Repository](#), [executor](#).

MechAgents supplies Colab notebooks, but its README specifies older FEniCS, AutoGen and OpenAI versions. Use it as a reproduction source with a separately recorded environment, rather than silently mixing its dependencies into this project. [Repository](#).

SciForma’s README describes a 9B diffusion backbone and an eight-B200 training setup, with other distributed configurations in its tree. A single Colab GPU would be a reduced adaptation experiment, not an equivalent reproduction of that training setup. [Repository](#).

7 Paper archive

The catalog and bibliography are in `papers/`. Downloaded PDFs retain their original authorship and publication notices. The download manifest distinguishes bundled PDFs from source links and unavailable downloads. This package does not bypass publisher access controls.